

QUBO Formulation: N-Queens Exact-Cover Problem

N-Queens Exact-Cover Problem

Problem Statement. Given an $N \times N$ chessboard, place N queens such that no two queens attack each other. This requirement is equivalent to an exact-cover formulation where exactly one queen must occupy each row, exactly one queen must occupy each column, and at most one queen may occupy each diagonal (both main and anti-diagonals). The goal is to find a feasible placement that satisfies all constraints simultaneously.

Decision Variables. Let $x_{i,j} \in \{0, 1\}$ for $i, j \in \{0, \dots, N-1\}$ be binary decision variables. The variable $x_{i,j}$ equals 1 if and only if a queen is placed at row i and column j , and 0 otherwise. There are N^2 variables in total, indexed by the board coordinates.

QUBO Derivation. Step 1. The problem is feasibility-driven, so we minimize the total number of violated constraints. This is achieved by minimizing the sum of squared constraint violations:

$$H(x) = \sum_{i=0}^{N-1} \left(\sum_{j=0}^{N-1} x_{i,j} - 1 \right)^2 + \sum_{j=0}^{N-1} \left(\sum_{i=0}^{N-1} x_{i,j} - 1 \right)^2 + \sum_d \left(\sum_{(i,j) \in d} x_{i,j} \right)^2, \quad (1)$$

where the first sum enforces row constraints, the second enforces column constraints, and the third enforces diagonal constraints.

Step 2. The original algebraic constraints are:

$$\sum_{j=0}^{N-1} x_{i,j} = 1, \quad \forall i \in \{0, \dots, N-1\}, \quad (2)$$

$$\sum_{i=0}^{N-1} x_{i,j} = 1, \quad \forall j \in \{0, \dots, N-1\}, \quad (3)$$

$$\sum_{(i,j) \in d} x_{i,j} \leq 1, \quad \forall \text{ diagonals } d. \quad (4)$$

Step 3. We convert each constraint to a quadratic penalty term. For a fixed row i , the squared penalty expands as:

$$\left(\sum_{j=0}^{N-1} x_{i,j} - 1 \right)^2 = \sum_{j=0}^{N-1} x_{i,j}^2 - 2 \sum_{j=0}^{N-1} x_{i,j} + 1. \quad (5)$$

Applying the idempotent property $x_{i,j}^2 = x_{i,j}$ for binary variables, this simplifies to $1 - \sum_{j=0}^{N-1} x_{i,j}$. By identical reasoning, the column penalty for a fixed j simplifies to $1 - \sum_{i=0}^{N-1} x_{i,j}$. For the diagonal penalty, expanding the square yields:

$$\left(\sum_{(i,j) \in d} x_{i,j} \right)^2 = \sum_{(i,j) \in d} x_{i,j}^2 + \sum_{\substack{(i,j), (k,l) \in d \\ (i,j) \neq (k,l)}} x_{i,j} x_{k,l} = \sum_{(i,j) \in d} x_{i,j} + \sum_{\substack{(i,j), (k,l) \in d \\ (i,j) \neq (k,l)}} x_{i,j} x_{k,l}. \quad (6)$$

Summing the linear terms across all row, column, and diagonal penalties gives:

$$\sum_{i=0}^{N-1} \left(- \sum_{j=0}^{N-1} x_{i,j} \right) + \sum_{j=0}^{N-1} \left(- \sum_{i=0}^{N-1} x_{i,j} \right) + \sum_d \sum_{(i,j) \in d} x_{i,j}. \quad (7)$$

Each cell (i, j) belongs to exactly two diagonals (one main and one anti-diagonal), so the diagonal linear sum equals $2 \sum_{i,j} x_{i,j}$. The total linear coefficient for every $x_{i,j}$ is therefore $-1 - 1 + 2 = 0$, meaning all linear terms cancel exactly. The constant terms sum to $\sum_i 1 + \sum_j 1 = 2N$.

Step 4. Combining the constant offset and the remaining quadratic terms, the complete QUBO Hamiltonian is:

$$H(x) = 2N + \sum_d \sum_{\substack{(i,j), (k,l) \in d \\ (i,j) \neq (k,l)}} 2x_{i,j} x_{k,l}. \quad (8)$$

Step 5. Reading off the QUBO parameters from the general closed form:

$$Q_{(i,j), (i,j)} = 0, \quad \forall i, j \in \{0, \dots, N-1\}, \quad (9)$$

$$Q_{(i,j), (k,l)} = \begin{cases} 2 & \text{if } (i,j) \text{ and } (k,l) \text{ share a diagonal,} \\ 0 & \text{otherwise,} \end{cases} \quad (10)$$

$$c = 2N. \quad (11)$$

Penalty Weight Justification. The formulation employs a unit penalty weight absorbed directly into the squared constraint terms. Because the linear terms identically cancel, the energy landscape is determined solely by the quadratic diagonal interactions. Any violation of a row, column, or diagonal constraint increases the energy by at least 2. Since the objective is strictly feasibility-driven, a penalty weight of 1 (yielding quadratic coefficients of 2) is sufficient to ensure that infeasible configurations always possess strictly higher energy than feasible ones, regardless of N .

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{N^2}$ recovers the optimal solution because feasible placements incur zero penalty from the constraint terms, yielding a constant energy $H(x) = 2N$. Any infeasible placement violates at least one constraint, resulting in a strictly positive penalty contribution and thus $H(x) > 2N$. Consequently, the global minimum of $H(x)$ corresponds exactly to a valid, non-attacking queen configuration, and the minimum energy value directly indicates the number of violated constraints (zero for feasible solutions).

Illustrative Example. Consider a concrete instance with $N = 4$. The board contains 16 cells, indexed by $k = 4i + j$ for $i, j \in \{0, 1, 2, 3\}$. The Hamiltonian becomes $H(x) = 8 + \sum_d \sum_{(i,j) \neq (k,l) \in d} 2x_{i,j}x_{k,l}$. The constant offset is $c = 8$. A valid configuration places queens at coordinates $(0, 1)$, $(1, 3)$, $(2, 0)$, and $(3, 2)$, corresponding to binary variables x_1, x_7, x_8, x_{14} set to 1, and all others to 0. Since no two selected cells share a row, column, or diagonal, all quadratic penalty terms vanish. The minimum energy is $H(x^*) = 8.0$, confirming a feasible solution with zero violations.